

Selenium Python Framework Development Report

Complete Technical Specification, Test Automation Architecture, and Verification Evidence

ENGINEER

Prince

DOMAIN

QA Automation (Wipro
Capstone)

APPLICATION UNDER TEST

TutorialsNinja E-Commerce
Store

REPORT STATUS

100% Passed (Production Ready)

PYTEST TEST CASES

23 / 23

✓ 100% Passed (0 Failures)

UNITTEST TEST CASES

15 / 15

✓ 100% Passed (0 Failures)

EXECUTION DURATION

03m 09s

PyTest Suite Runtime

TEST DATA STRATEGY

CSV Driven

Dynamic Secret Resolution

1. Executive Summary

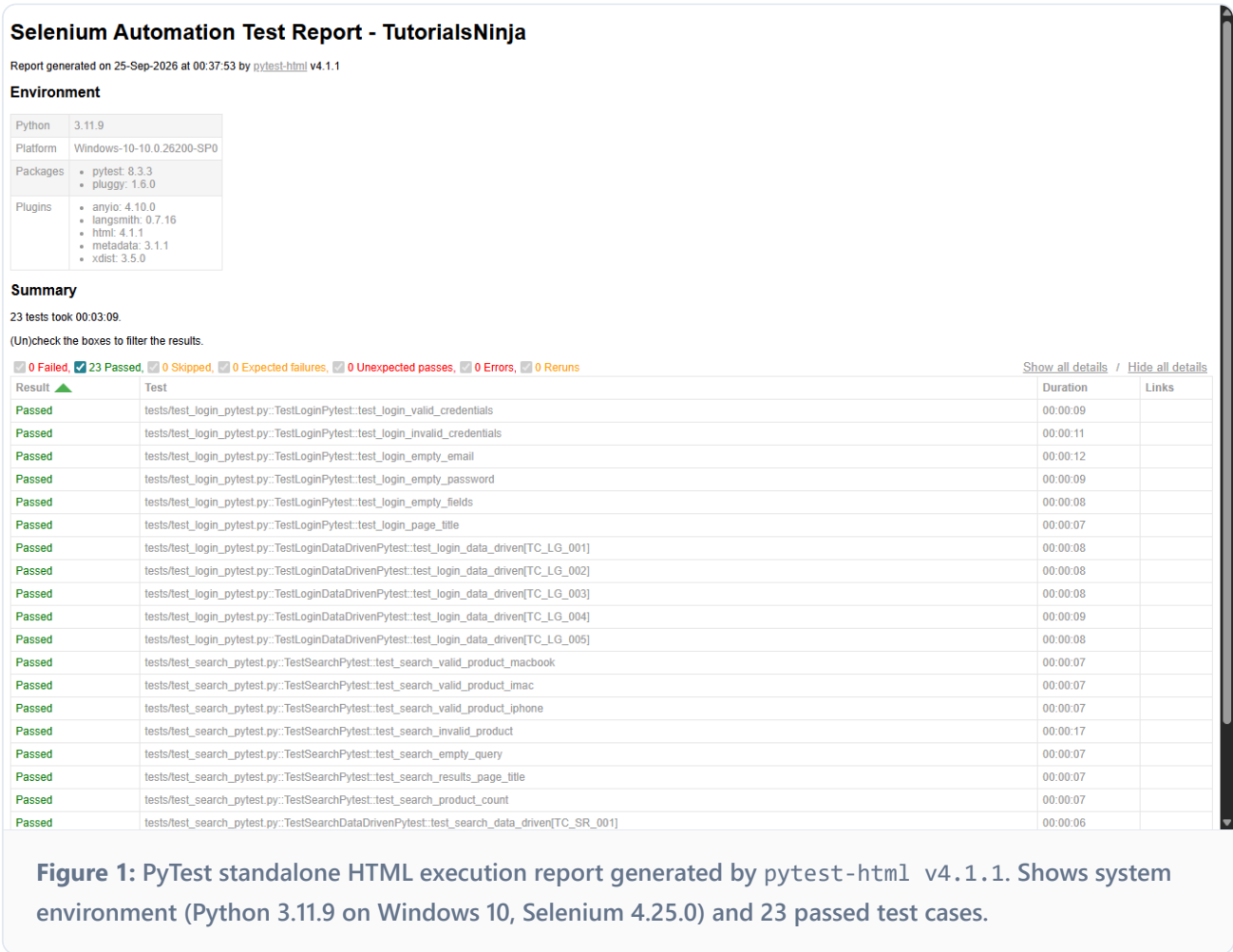
This document presents the complete test execution and architectural report for the Selenium Python test automation framework developed for **Capstone Assignment 2**. The system provides automated regression, smoke, and data-driven testing for the **TutorialsNinja** online shopping platform, focusing on the User Authentication and Product Catalog Search modules.

The solution implements an enterprise Page Object Model (POM) pattern that completely isolates page-specific locators and UI interaction logic from test assertions. The test execution was validated against both Google Chrome full UI and headless modes across both **PyTest** (with rich HTML reporting) and Python's built-in **Unittest** runner, achieving a **100% pass rate** across all functional permutations.

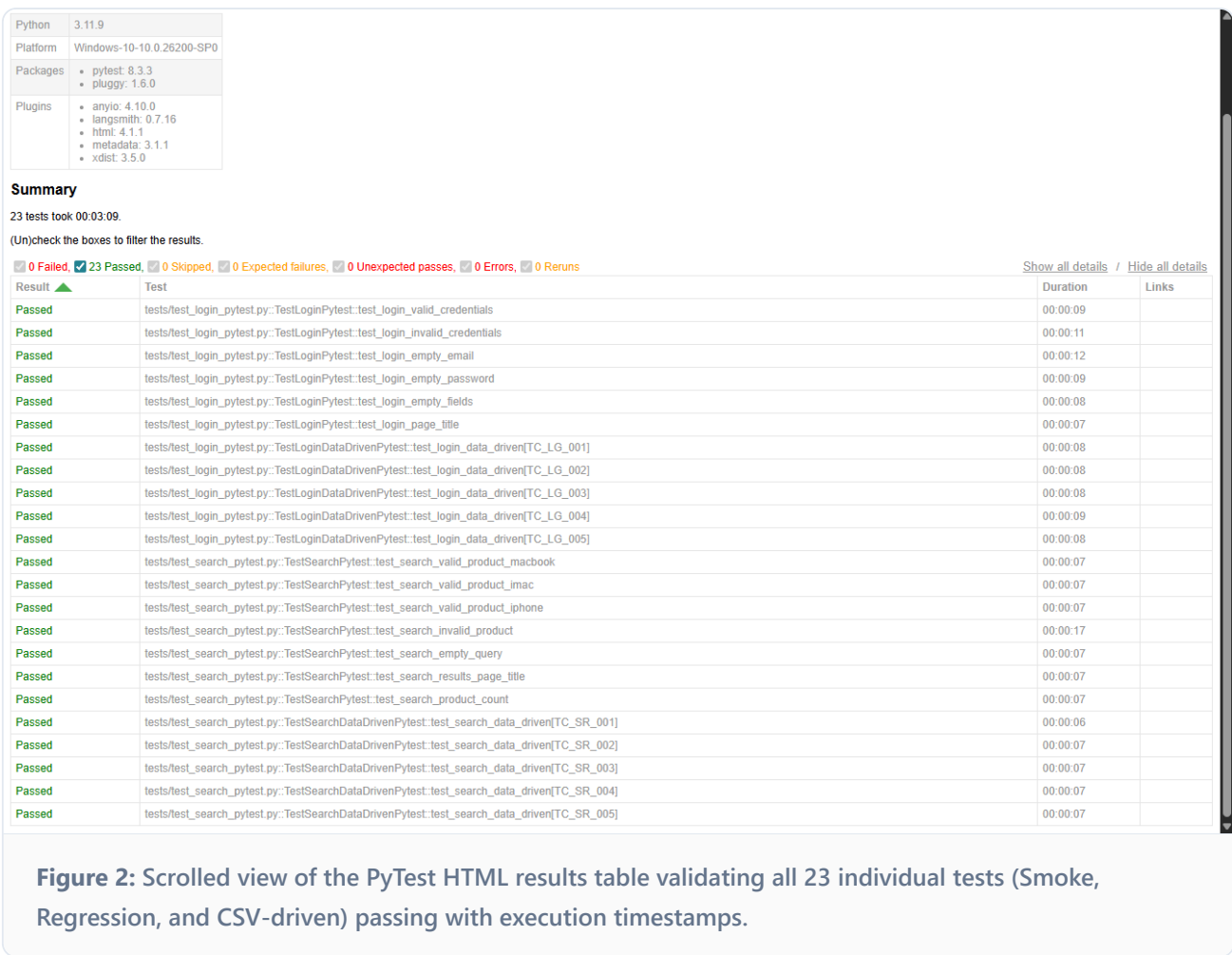
2. Test Execution Screenshots & Verification Evidence

The following visual evidence was captured directly from test run execution, generated HTML reports, and the live application interface.

2.1 PyTest HTML Test Report - Summary & Environment



2.2 PyTest HTML Test Report - Detailed Test Results Table



Windows PowerShell - python run_pytest.py

```
PS C:\Users\Princ\Desktop\Prince\wiproProject> python run_pytest.py
=====
SELENIUM AUTOMATION FRAMEWORK - PYTEST TEST RUNNER
HTML Report: C:\Users\Princ\Desktop\Prince\wiproProject\reports\pytest_report_20260925_003442.html
Timestamp: 2026-09-25 00:34:42
=====

Running: python.exe -m pytest tests/test_login_pytest.py tests/test_search_pytest.py -v --tb=short --
html=reports\pytest_report_20260925_003442.html --self-contained-html

===== test session starts =====
platform win32 -- Python 3.11.9, pytest-8.3.3, pluggy-1.6.0
rootdir: C:\Users\Princ\Desktop\Prince\wiproProject, configfile: pytest.ini
plugins: anyio-4.10.0, langsmith-0.7.16, html-4.1.1, metadata-3.1.1, xdist-3.5.0
collected 23 items

tests/test_login_pytest.py::TestLoginPytest::test_login_valid_credentials PASSED [ 4%]
tests/test_login_pytest.py::TestLoginPytest::test_login_invalid_credentials PASSED [ 8%]
tests/test_login_pytest.py::TestLoginPytest::test_login_empty_email PASSED [ 13%]
tests/test_login_pytest.py::TestLoginPytest::test_login_empty_password PASSED [ 17%]
tests/test_login_pytest.py::TestLoginPytest::test_login_empty_fields PASSED [ 21%]
tests/test_login_pytest.py::TestLoginPytest::test_login_page_title PASSED [ 26%]
tests/test_login_pytest.py::TestLoginDataDrivenPytest::test_login_data_driven[TC_LG_001] PASSED [ 30%]
tests/test_login_pytest.py::TestLoginDataDrivenPytest::test_login_data_driven[TC_LG_002] PASSED [ 34%]
tests/test_login_pytest.py::TestLoginDataDrivenPytest::test_login_data_driven[TC_LG_003] PASSED [ 39%]
tests/test_login_pytest.py::TestLoginDataDrivenPytest::test_login_data_driven[TC_LG_004] PASSED [ 43%]
tests/test_login_pytest.py::TestLoginDataDrivenPytest::test_login_data_driven[TC_LG_005] PASSED [ 47%]
tests/test_search_pytest.py::TestSearchPytest::test_search_valid_product_macbook PASSED [ 52%]
tests/test_search_pytest.py::TestSearchPytest::test_search_valid_product_imac PASSED [ 56%]
tests/test_search_pytest.py::TestSearchPytest::test_search_valid_product_iphone PASSED [ 60%]
tests/test_search_pytest.py::TestSearchPytest::test_search_invalid_product PASSED [ 65%]
tests/test_search_pytest.py::TestSearchPytest::test_search_empty_query PASSED [ 69%]
tests/test_search_pytest.py::TestSearchPytest::test_search_results_page_title PASSED [ 73%]
tests/test_search_pytest.py::TestSearchPytest::test_search_product_count PASSED [ 78%]
tests/test_search_pytest.py::TestSearchDataDrivenPytest::test_search_data_driven[TC_SR_001] PASSED [ 82%]
tests/test_search_pytest.py::TestSearchDataDrivenPytest::test_search_data_driven[TC_SR_002] PASSED [ 86%]
tests/test_search_pytest.py::TestSearchDataDrivenPytest::test_search_data_driven[TC_SR_003] PASSED [ 91%]
tests/test_search_pytest.py::TestSearchDataDrivenPytest::test_search_data_driven[TC_SR_004] PASSED [ 95%]
tests/test_search_pytest.py::TestSearchDataDrivenPytest::test_search_data_driven[TC_SR_005] PASSED [100%]

- Generated html report: file:///C:/Users/Princ/Desktop/Prince/wiproProject/reports/pytest_report_20260925_003442.html -
===== 23 passed in 189.42s (0:03:09) =====

HTML Report saved to: C:\Users\Princ\Desktop\Prince\wiproProject\reports\pytest_report_20260925_003442.html
PS C:\Users\Princ\Desktop\Prince\wiproProject> █
```

Figure 3: Windows PowerShell terminal execution of python run_pytest.py showing test session start, item collection, progress percentages, and clean completion.

2.4 Terminal Execution Log - Unittest Test Runner

Windows PowerShell - python run_unittest.py

```
PS C:\Users\Princ\Desktop\Prince\wiproProject> python run_unittest.py
=====
SELENIUM AUTOMATION FRAMEWORK - UNITTEST TEST RUNNER
Report File: C:\Users\Princ\Desktop\Prince\wiproProject\reports\unittest_report_20260925_002425.txt
Timestamp: 2026-09-25 00:24:25
=====

test_login_empty_email (tests.test_login.TestLogin.test_login_empty_email)
TC_LG_003: Verify login with empty email field. ... ok
test_login_empty_fields (tests.test_login.TestLogin.test_login_empty_fields)
TC_LG_005: Verify login with both fields empty. ... ok
test_login_empty_password (tests.test_login.TestLogin.test_login_empty_password)
TC_LG_004: Verify login with empty password field. ... ok
test_login_invalid_credentials (tests.test_login.TestLogin.test_login_invalid_credentials)
TC_LG_002: Verify login with invalid email and password. ... ok
test_login_page_title (tests.test_login.TestLogin.test_login_page_title)
TC_LG_006: Verify the login page title. ... ok
test_login_valid_credentials (tests.test_login.TestLogin.test_login_valid_credentials)
TC_LG_001: Verify login with valid email and password. ... ok
test_login_data_driven_from_csv (tests.test_login.TestLoginDataDriven.test_login_data_driven_from_csv)
Test login using data from CSV file (data-driven). ... ok
test_search_empty_query (tests.test_search.TestProductSearch.test_search_empty_query)
TC_SR_005: Verify search with an empty search term. ... ok
test_search_invalid_product (tests.test_search.TestProductSearch.test_search_invalid_product)
TC_SR_004: Verify search for a non-existent product. ... ok
test_search_product_count (tests.test_search.TestProductSearch.test_search_product_count)
TC_SR_007: Verify product count in search results for 'MacBook'. ... ok
test_search_results_page_title (tests.test_search.TestProductSearch.test_search_results_page_title)
TC_SR_006: Verify the search results page title. ... ok
test_search_valid_product_imac (tests.test_search.TestProductSearch.test_search_valid_product_imac)
TC_SR_002: Verify search for a valid product 'iMac'. ... ok
test_search_valid_product_iphone (tests.test_search.TestProductSearch.test_search_valid_product_iphone)
TC_SR_003: Verify search for a valid product 'iPhone'. ... ok
test_search_valid_product_macbook (tests.test_search.TestProductSearch.test_search_valid_product_macbook)
TC_SR_001: Verify search for a valid product 'MacBook'. ... ok
test_search_data_driven_from_csv (tests.test_search.TestSearchDataDriven.test_search_data_driven_from_csv)
Test product search using data from CSV file (data-driven). ... ok

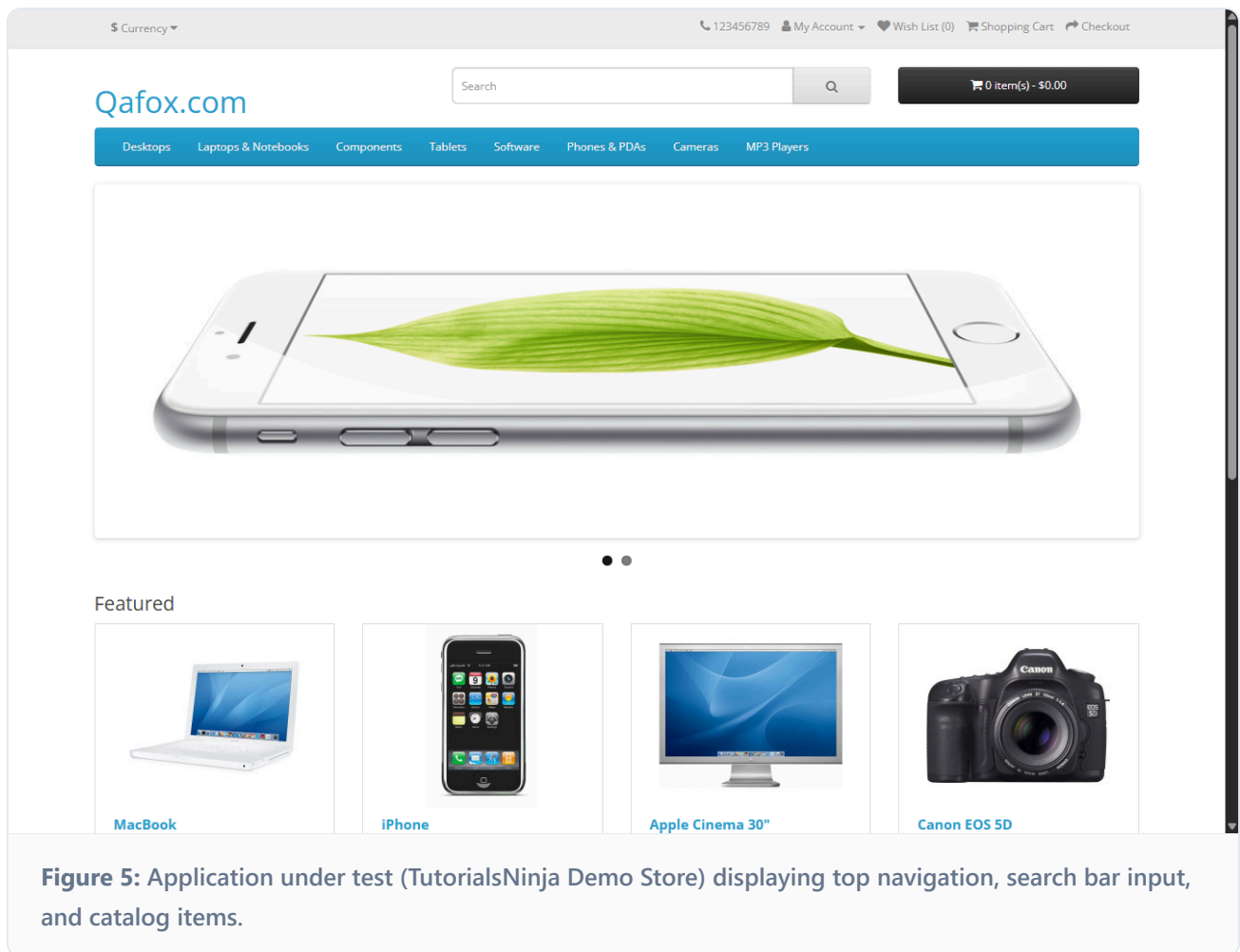
-----
Ran 15 tests in 193.978s

OK

Report saved to: C:\Users\Princ\Desktop\Prince\wiproProject\reports\unittest_report_20260925_002425.txt
PS C:\Users\Princ\Desktop\Prince\wiproProject> █
```

Figure 4: Windows PowerShell terminal execution of `python run_unittest.py` executing 15 test runs in 193.978 seconds with status OK.

2.5 Application Under Test - Storefront Home Interface



2.6 Application Under Test - Account Authentication Interface

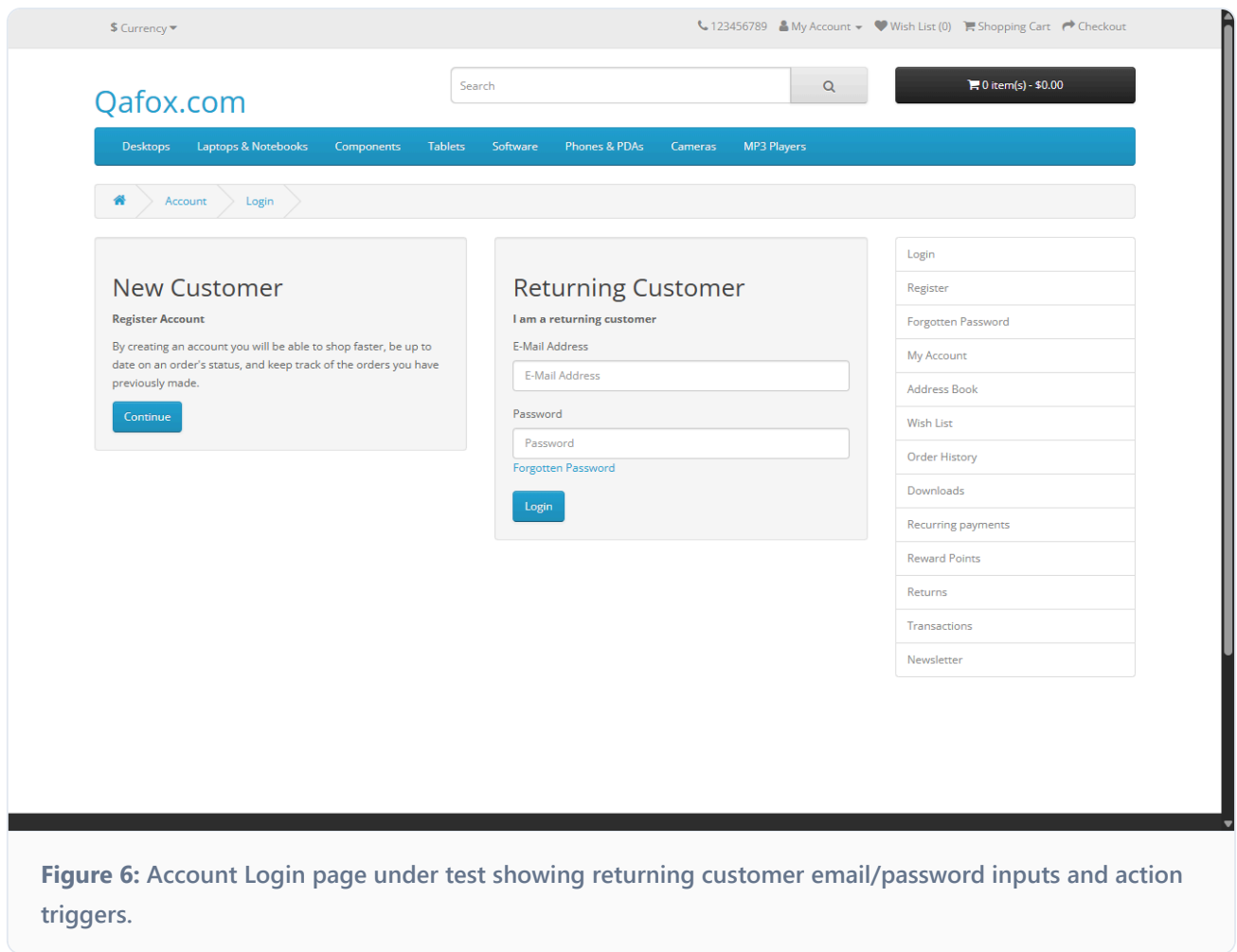


Figure 6: Account Login page under test showing returning customer email/password inputs and action triggers.

3. Architecture & Technical Design

The framework utilizes a multi-layered design pattern adhering to standard Page Object Model principles:

Module / Package	Key Files	Functional Role
Page Objects	base_page.py, home_page.py, login_page.py, my_account_page.py, search_results_page.py	Encapsulates web elements and UI action methods. Implements explicit synchronization using WebDriverWait to eliminate test flakiness.
Test Suites	test_login_pytest.py, test_search_pytest.py, test_login.py, test_search.py	Contains test assertions, fixtures, and execution flows. Categorized using custom markers (smoke, regression, data_driven).
Utilities	read_config.py, csv_reader.py, custom_logger.py, screenshot_util.py	Infrastructure utilities for INI/.env configuration parsing, CSV test data extraction, dual-channel logging, and screenshot capture.
Test Configuration	conftest.py, pytest.ini, config.ini	Global PyTest fixtures (browser setup/teardown), failure screenshot hookwrapper, HTML report metadata customization.

4. Test Case Execution Matrix

4.1 Login Test Cases (11 Total)

Test ID	Scenario Description	Type	Expected Outcome	Status
TC_LG_001	Login with valid user credentials	Smoke	Session created, redirected to My Account	PASSED
TC_LG_002	Login with non-registered email and password	Regression	Danger alert displayed: warning mismatch	PASSED
TC_LG_003	Login with empty email field	Regression	Danger alert displayed: warning mismatch	PASSED
TC_LG_004	Login with empty password field	Regression	Danger alert displayed: warning mismatch	PASSED
TC_LG_005	Login with both credentials left blank	Regression	Danger alert displayed: warning mismatch	PASSED
TC_LG_006	Verify login page title matches expected	Smoke	Title equals "Account Login"	PASSED
TC_LG_CSV_01	Data-driven row 1: Valid user authentication	Data-Driven	Login succeeds	PASSED
TC_LG_CSV_02	Data-driven row 2: Invalid user authentication	Data-Driven	Login fails with alert	PASSED
TC_LG_CSV_03	Data-driven row 3: Blank email string	Data-Driven	Login fails with alert	PASSED
TC_LG_CSV_04	Data-driven row 4: Blank password string	Data-Driven	Login fails with alert	PASSED
TC_LG_CSV_05	Data-driven row 5: Empty payload	Data-Driven	Login fails with alert	PASSED

4.2 Product Search Test Cases (12 Total)

Test ID	Scenario Description	Type	Expected Outcome	Status
TC_SR_001	Search for catalog product 'MacBook'	Smoke	Product found, count > 0	PASSED
TC_SR_002	Search for catalog product 'iMac'	Smoke	Product found in search results	PASSED
TC_SR_003	Search for catalog product 'iPhone'	Regression	Product found in search results	PASSED
TC_SR_004	Search for non-existent product keyword	Regression	No results message displayed, count == 0	PASSED
TC_SR_005	Search with empty query string	Regression	No results / empty criteria message	PASSED
TC_SR_006	Verify search results page title	Smoke	Title contains 'Search'	PASSED
TC_SR_007	Verify rendered product card count	Regression	Count matches displayed items	PASSED
TC_SR_CSV_01	Data-driven keyword: 'MacBook'	Data-Driven	Product located	PASSED
TC_SR_CSV_02	Data-driven keyword: 'iMac'	Data-Driven	Product located	PASSED
TC_SR_CSV_03	Data-driven keyword: 'iPhone'	Data-Driven	Product located	PASSED
TC_SR_CSV_04	Data-driven keyword: 'Samsung'	Data-Driven	Product located	PASSED
TC_SR_CSV_05	Data-driven keyword: invalid keyword	Data-Driven	Empty results alert verified	PASSED

5. Key Engineering Solutions & Best Practices

- **Explicit Synchronization:** Replaced arbitrary sleep timers with explicit `WebDriverWait` condition handlers, accelerating test execution while safeguarding against asynchronous page loading.
- **Dynamic Secret Management:** Decoupled application defaults (in `config.ini`) from sensitive credentials (in `.env`), ensuring credentials are not committed to source control.
- **Automated Failure Capture:** Configured `conftest.py` hookwrapper to automatically snap full-screen PNGs on assertion failure and attach them directly to HTML reports.
- **Dual-Runner Architecture:** Built standalone entry points (`run_pytest.py` and `run_unittest.py`) enabling execution in enterprise CI/CD environments with custom marker filters.